13 December 1982

SUPERSEDING
MIL-R-55317(EL)
10 November 1970

MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/URR-70 AND KIT, ACCESSORY MK-1517/UR

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification defines the requirements for the following units of a general purpose high frequency superheterodyne receiver and an associated accessory kit:

Receiving Set, Radio AN/URR-70, consisting of:
Receiver, Radio R-1218/UR
Antenna, Whip AS-2887/UR
Field Pack, Canvas CW-1005/UR

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

1.1 Scope. (Cont'd)

Kit, Accessory MK-1517/UR, consisting of:
Base, Shock Mount, Electrical Equipment MT-4034/UR
Cable Assembly, Special Purpose CX-10956/U (10 ft. 0 in.)
Cable Assembly, Special Purpose CX-10957/U (10 ft. 0 in.)
Cable Assembly, Power, Electrical CX-10958/U (10 ft. 0 in.)
Cable Assembly, Extension CX-12953/U (2 ft. 0 in.)
Field Pack, Canvas CW-1005/UR

1.2 Description. The equipment consists of Radio Receiver R-1218/UR and Accessory Kit MK-1517/UR, described as follows:

1.2.1 Radio Receiver, R-1218/UR. The basic Radio Receiver R-1218 is a superheterodyne receiver capable of receiving AM, CW, and SSB signals, in five bands, over a frequency range of 0.536 to 20.5 MHz. The receiver is powered by 115/230 V, 47 to 400 Hz AC, 26 Vdc vehicular, or an internally contained dry battery power source, consisting of 12 type BA-30/U or BA-303/U cells. The basic receiver package also includes a whip antenna AS-2887/UR and canvas Field Pack CW-1005/UR.

1.2.2 Accessory Kit MK-1517/UR. The Accessory Kit MK-1517/UR consists of four cables; 115 Vac power cable, CX-10956/U (10 ft. 0 in.), 230 Vac power cable, CX 10957/U (10 ft. 0 in.), 26 Vdc power cable, CX-10958/U (10 ft. 0 in.) and an extension cable, CX-12953/U (2 ft. 0 in.); a Shock Mount Base, MT-4034/UR, and a Canvas Field Pack CW-1005/UR to contain the kit items.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents form a part of this specification to the extent specified herein. The effective issue or revision of specifications and standards shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (a) specific issues are set forth herein, or (b) different issues than those specified in this specification are set forth in the solicitation. The date of the applicable DODISS and supplements thereto, and the effective issue or revision of drawings cited herein, shall be as specified in the solicitation or contract.

SPECIFICATIONS

FEDERAL

CCC-C-419 Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Cont'd)

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-3885B	Cable Assemblies and Cord Assemblies, Electrical
MIL-L-3891	Luminescent Material and Equipment (Nonradioactive)
MIL-R-10273	Radio Sets, AN/PRC-8( ), AN/PRC-9( ), AN/PRC-10( )
MIL-P-11268K	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231(EL)	Marking of Electronic Items
MIL-F-14072(EL)	Finishes for Ground Electronic Equipment
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for. Also include supplements /21 and/31

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A w/Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 w/Notice 3	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28 Volt Dc Transient Characteristics for Military Vehicles
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

DRAWINGS

U.S. Army Electronics Research and Development Command

DL-SC-B-91304	Antenna, AT-272( )/PRC
SC-A-678020	Burn-In Procedure for Radio Receivers, R-1218/UR and R-1518/UR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Cont'd)

DRAWINGS

SM-A-747531	Test Procedure-Receiver, Radio, R-1218/UR
DL-SM-B-747551	Antenna, AS-2887/UR
DL-SM-B-747552	Kit, Accessory, MK-1517/UR
DL-SM-B-747557	Base, Shock Mount, Electrical Equipment, MT-4034/UR
DL-SM-B-747558	Field Pack, Canvas, CW-1005/UR
DL-SM-B-747559	Receiving Set, Radio, AN/URR-70
DL-SM-B-747564	Maintenance Item Set
SM-B-747787	Lamp, Incandescent, Modified

SPECIAL INSTRUCTIONS

SPI 1G00075	Special Packaging Instructions (see para. 5.1).

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS.

3.1 Basic receiver complement. The AN/URR-70 receiving set consists of the following listed items only, and requires additional items to supply audio response and power (see 3.2 below).

a. Receiver, Radio R-1218/UR
b. Antenna, Whip AS-2887/UR
c. Field Pack, Canvas CW-1005/UR
d. Running spare parts in pocket of field pack cover, including:

(1) Fuses, cartridge, 4 ea., Part No. F02A250V1-8A
(2) Lamp, Incandescent, Modified, 4 ea., Part No. SM-B-747787

3.2 Ancillary items. The following ancillary items are not part of the basic receiver complement, but are used with it. They will be procured and used on an "as required" basis.

3.2.1 Kit, Accessory, MK-1517/UR. The accessory kit, supplied in accordance with DL-SM-B-747552, includes one each of the following items:

3.3.1.1 Field Pack, Canvas, CW-1005/UR. A water repellant, cotton duck, zipper closing, carrying case meeting the requirements of 3.5.1.3 and DL-SM-B-747558 shall be supplied with each accessory kit. It shall contain the 3 power cables, the extension cable and the shock mount base. An identical canvas field pack shall be supplied to house the R-1218/UR receiver, whip antenna and running spares.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.2.1.2 Base, Shock Mount, Electrical Equipment, MT-4034/UR. The shock mount base, provided to protect the receiver from damage during vehicular operation, shall be made in accordance with DL-SM-B-747557.

3.2.1.3 Cables. The 4 cables listed below shall be part of the accessory kit. They shall be constructed in accordance with applicable drawings of DL-SM-B-747552 and shall meet specific requirements as outlined below:

a. Cable Assembly, Special Purpose CX-10956/U (10 ft. 0 in.). The 115 volt AC Power Cable shall be type CO-03 MGF (3/18) 0340 conforming to MIL-C-3432 and Paragraphs 3.1, 3.3, 3.3.3, 3.8, 3.9/4.6.1 (Visual and Mechanical), 3.7.3/4.6.3.3 (Twist and pull with continuity as post operational test), 3.6.1/4.6.2.1 (Continuity), 3.6.3/4.6.2.3 (Dielectric withstanding voltage), and 3.6.4./4.6.2.4 (Insulation Resistance) of MIL-C-3885B.

b. Cable Assembly, Special Purpose CX-10957/U (10 ft. 0 in.). The 230 volt AC power cable shall be type CO-03MGF (3/18) 0340 conforming to MIL-C-3432 and paragraphs 3.1, 3.3, 3.3.3, 3.8, 3.9/4.6.1 (Visual and Mechanical), 3.7.3/4.6.3.3 (Twist and pull with continuity as post operational test), 3.6.1/4.6.2.1 (Continuity), 3.6.3/4.6.2.3 (Dielectric withstanding voltage), and 3.6.4/4.6.2.4 (Insulation Resistance) of MIL-C-3885B.

c. Cable Assembly, Special Purpose CX-10958/U (10 ft. 0 in.). The 26 volt DC, vehicular power cable shall be type CO-02MGF (2/18) 0310 conforming to MIL-C-3432 and paragraphs 3.1, 3.3, 3.3.3, 3.8, 3.9/4.6.1 (Visual and Mechanical), 3.7.3/4.6.3.3 (Twist and pull with continuity as post operational test), 3.6.1/4.6.2.1 (Continuity), 3.6.3/4.6.2.3 (Dielectric withstanding voltage), and 3.6.4/4.6.2.4 (Insulation Resistance) of MIL-C-3885B.

d. Cable Assembly, Extension CX-12953/UR (2 ft. 0 in.). The extension cable assembly, used to operate the receiver out of its case, shall meet the requirements from the following paragraphs of MIL-C-3885B:

3.1, 3.3, 3.3.3, 3.8, 3.9/4.6.1 (Visual and Mechanical)
3.7.3/4.6.3.3 (Twist and pull with continuity as post operational test)
3.6.1/4.6.2.1 (Continuity)
3.6.3/4.6.2.3 (Dielectric withstanding voltage)
3.6.4/4.6.2.4 (Insulation Resistance)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.2.2 Maintenance Item Set. The Maintenance Item Set contains those special purpose extender cables, tuning tools, connectors, etc. required for DS/GS level maintenance of the R-1218/UR receiver, which capability is not available as standard test equipment. The set shall be supplied in accordance with DL-SM-B-747564.

3.2.3 Audio output device. The Receiving Set AN/URR-70 does not include earphones or a speaker as an audio output transducer, thus they are not a concern of this specification. The audio output of the receiving set is a 600 ohm termination.

3.2.4 Internal batteries. As well as being powered by 115/230 VAC and 26 VDC, the R-1218/UR receiver may be powered by internal dry cells. The battery source is 18 VDC nominal, supplied by 12 each BA-30/U or BA-3030/U dry cells.

3.3 Burn-in. Each receiver to be delivered on contract, or to be subjected to first article tests or reliability tests in accordance with this specification, shall be subjected to a 48 hour burn-in to limit infant mortality. The last 24 hour burn-in period shall be failure free. This burn-in shall be prior to any or all tests specified herein (see 4.3).

3.4 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.4 (see 6.2.d and 6.2.f).

3.5 Mechanical and construction requirements. The equipment shall be constructed in accordance with the requirements of this specification and drawings DL-SM-B-747559, DL-SM-B-747552 and DL-SM-B-747564 which describe the Receiving Set, AN/URR-70, the Accessory Kit, MK-1517/UR and the Maintenance Item Set, including the requirements for parts, materials and processes thereon.

3.5.1 Parts, materials and processes; general. In addition to the requirements of this specification, the general requirements of MIL-P-11268K(Army), including the selection requirements therein, shall apply.

3.5.1.1 Finish. The equipment shall be finished in accordance with MIL-F-14072(EL) and the equipment drawings. The final paint film on all exterior surfaces shall be green color (olive drab) semigloss enamel matching a color chip provided by the procuring agency (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.5.1.2 Marking.

3.5.1.2.1 General. Marking shall conform to MIL-M-13231(EL) (see 4.5). Front panel marking shall be Group I as described in MIL-M-13231(EL). Location of marking shall be as specified on the equipment drawings.

3.5.1.2.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.1.2.3 Luminescent marking. Luminescent marking shall be type P in accordance with Specification MIL-L-3891.

3.5.1.2.4 Serial numbers. Each Radio Receiver R-1218/UR shall be serial numbered. The format of the serial number and the starting number will be specified in the procurement document.

3.5.1.3 Fabric and thread.

3.5.1.3.1 General. The color of fabric shall be Olive-Drab No. 7. This requirement shall be met immediately after treatment. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread (see 6.5). Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature. Specification and treatment of materials shall be in accordance with paragraphs 3.8.3 and 3.9.7 of MIL-P-11268K(Army).

3.5.1.3.2 Duck. Cotton duck shall be in accordance with Federal Specification CCC-C-419, Type 1, Number 6 duck.

3.5.1.3.3 Stitches, seams, and stitching. Stitches, seams, and stitching shall conform to Federal Standard FED-STD-751.

3.5.1.4 Tropicalization (fungus-inert materials). No fungus nutrient materials, treated or otherwise, shall be used except in a hermetically sealed enclosure for the Receiver R-1218/UR. Fungus-inert materials shall be in accordance with requirement 4 of MIL-STD-454. (see 3.7.9 and 4.9.14)

3.5.2 Size and weight. The size of the receiver shall be 5 ± 1/16 inches high by 7 1/8 ± 1/16 inches wide x 9 3/8 ± 1/16 inches long. The weight shall not exceed 11 pounds (without batteries).

3.5.3 Tuning control. Two speed tuning shall scan each entire bandwidth in approximately 12 revolutions at the fast tuning rate. The slow tuning rate shall be one-fourtieth of the fast tuning rate ± 20%. The mechanical stop at each end of the tuning range shall be capable of withstanding 10 inch pounds of torque.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.5.4 Whip antenna, AS-2887/UR. The whip antenna shall be constructed to and meet the requirements of DL-SM-B-747551. It is made from Antenna AT-272( )/PRC, conforming to Specification MIL-R-10273 and drawing number DL-SC-B-91304.

3.5.5 Equipment safety engineering.

3.5.5.1 Personnel hazards. Personnel hazards shall be kept to a minimum by control of configuration changes and parts selection to assure compliance with Requirement 1 of MIL-STD-454. Compliance with these requirements shall be verified through a visual inspection (see 4.7).

3.5.5.2 Edge rounding, exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm) and exposed corners to a minimum of 0.03 inch (0.75 mm).

3.5.6 Human engineering. Configuration design criteria affecting human factors operation, maintainability and handling shall be maintained through compliance with MIL-STD-1472.

3.5.7 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the system and shall conform to Requirement 7 of MIL-STD-454 (see 4.10). Individual items shall not be selected for fit or performance, however, matched pairs or sets when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating or characteristic.

3.6 Electrical requirements.

3.6.1 Antenna inputs.

3.6.1.1 Input impedance. The ANT 2 antenna input terminal shall match a nominal 50 ohm impedance and the ATN 1 terminal shall match a high impedance affixable whip. The antenna selector switch shall select the antenna input for use and shall ground the unused connector.

3.6.1.2 Antenna input isolation. The isolation between the antenna inputs shall be greater than 60 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.2 Frequency range (overlap). The frequency range of the receiver shall be 0.536 to 20.5 MHz, covered in 5 bands as follows:

Band	Frequency range (MHz)
1	0.536 to 1.16
2	1.10 to 2.37
3	2.26 to 4.87
4	4.63 to 10.00
5	9.50 to 20.50

3.6.3 Frequency stability.

3.6.3.1 Frequency stability vs. time. When operating from a cold start, the stability of the high frequency oscillator shall not deviate more than 0.015% in the first 5 minutes nor more than 0.033% from 5 to 60 minutes. The short-term drift for any five-minute period of time shall not exceed 400 Hz or 0.004% whichever is greater, after a 5 minute warm-up.

3.6.3.2 Frequency stability vs. internal battery voltage. As the battery supply voltage is varied from 18 V to 14 V, the receiver frequency for any given dial setting shall not deviate more than ± 0.005%.

3.6.3.3 Frequency stability vs. received signal level. The observed frequency of maximum response, when receiving a signal of 0.5 V at the antenna input, shall not deviate more than 800 Hz from that observed at minimum sensitivity levels.

3.6.3.4 Frequency stability vs. temperature. When tested in accordance with 4.8.2/4.9.1, the frequency drift of the receiver shall not exceed 0.75% over the operating temperature range. The rate of frequency drift shall not exceed 0.006% per degree F change in temperature.

3.6.3.5 Frequency stability vs. humidity. The frequency drift of the receiver shall not exceed 0.125% throughout the humidity cycle when tested in accordance with 4.8.2/4.9.2.

3.6.4 Sensitivity.

3.6.4.1 AM and CW sensitivity. The signal level required to yield a 10 db signal plus noise to noise ratio, at an output reference level of 10 mW watts, shall not exceed 1.2 μV when a 30%, 1000 Hz amplitude modulated signal is impressed on the 50 ohm antenna input. CW sensitivity for the same signal plus noise to noise ratio and reference level shall not exceed 0.4 μV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.4.2 AM sensitivity (whip antenna input). The receiver shall have a 10db or better signal-plus-noise-to-noise ratio, with an output reference level of 10 mW when the following 1000 Hz, 30% modulated, AM signal levels are impressed on the whip antenna input under the test conditions of 4.8.2.

Band	Frequency (MHz)	Signal Level, uV
1	0.536	70
1	0.848	42
1	1.16	30
2	1.10	32
2	1.73	22
2	2.37	18
3	2.26	18
3	3.56	12
3	4.87	10
4	4.63	9
4	7.50	8
4	10.0	8
5	9.50	8
5	15.0	8
5	20.5	10

3.6.4.3 AM sensitivity vs. internal battery voltage. AM sensitivity of the receiver, when tested under the conditions of 3.6.4.1 above, shall not require more than a 2.0 uV signal input level to maintain a 10 db signal-plus-noise-to-noise ratio, with a 10 milliwatt reference output, when the battery supply voltage is reduced from 18 V to 14 V.

3.6.4.4 AM and CW sensitivity vs. temperature. At the operating temperature extremes, -40° and +154°F, the signal level required to produce a signal-plus-noise-to-noise ratio of 10 db, with an output reference level of 10 mW, shall not exceed 2.0 uV for AM operation and 0.7 uV for CW operation.

3.6.4.5 AM and CW sensitivity vs. humidity. Throughout the humidity cycle, the signal level required to produce a signal-plus-noise-to-noise ratio of 10 db, with an output reference level of 10 mW, shall not exceed 2.0 uV for AM operation and 0.7 uV for CW operation.

3.6.5 Calibration oscillator.

3.6.5.1 Frequency calibrator accuracy. The calibration oscillator frequency shall be 100 kHz ± 0.002% for use in the 0.536 to 4.87 MHz frequency range and 500 kHz ± 0.002% for use from 4.63 to 20.5 MHz. The calibration oscillator frequency change shall be accomplished by means of the band switch.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:16Z
Check the source to verify that this is the currenversion before use.

3.6.5.2 Frequency calibrator degradation. The calibration oscillator frequency shall not vary more than ± 0.01% under the various environmental conditions when tested in accordance with 4.8.2.

3.6.5.3 Injection level. The minimum permissible calibration oscillator injection shall be sufficient to yield a half scale deflection on the panel meter when the RF gain control is at maximum, while the maximum permissible injection shall not cause more than a half scale meter deflection when the RF gain control is set at minimum.

3.6.5.4 Spurious indications. There shall be no observable panel meter readings as the result of spurious calibration oscillator emissions when observed at maximum RF gain.

3.6.5.5 Calibration adjustment. The calibration cursor shall be capable of being adjusted by means of the CAL ADJ Control, both above and below the Calibration Frequency Mark on the dial at each calibration point in all five bands.

3.6.6 Dial readout error. Calibration adjustment at any check point shall result in no more than ± 3.0 kHz or 0.2% (whichever is larger) readout error at any point between that check point and adjacent calibration points.

3.6.7 Tuning backlash. The receiver tuning shall be directly calibrated in frequency. The control shall operate free from backlash and have a resetability of 0.1 percent or 2.5 kHz whichever is greater.

3.6.8 Spurious Responses.

3.6.8.1 Image rejection. Image rejection shall be supppressed at least 80 db from 0.536 to 3.0 MHz and 50 db from 3.0 to 20.5 MHz.

3.6.8.2 IF rejection. IF rejection shall be suppressed at least 50 db from 0.536 to 3.0 MHz and 80 db from 3.0 to 20.5 MHz.

3.6.8.3 Crossmodulation. Crossmodulation response shall be suppressed by the following amounts:

Generator 1 Reference Signal	Separation between desired and undesired signals	Generator 2 Signal Level
3 MHz 1.2 uV	± 20 kHz	0.12 mV(40 db down)
3 MHz 1.2 uV	± 50 kHz	1.2 mV (60 db down)
3 MHz 1.2 uV	± 125 kHz	12 mv(80 db down)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.8.4 Intermodulation distortion products. The intermodulation distortion products shall be suppressed at least 66 db except that in-band third order distortion products shall be suppressed at least 60 db.

3.6.9 Intermediate frequency amplifier.

3.6.9.1 IF center frequency. The IF center frequency shall be 455 kHz ± 0.1%.

3.6.9.2 IF selectivity. The receiver IF selectivity shall be as specified below:

IF Response Characteristics

Response (db down from max)	Input Level (microvolts)	Bandwidth (kilohertz)
reference	2.5	-----
6	5.0	2.5 to 3.5
60	2500	less than 10
70	7900	less than 16

3.6.10 AGC characteristics. The audio output shall not vary more than 6 db when signal levels from 2.5 µV to 0.1 V are applied to the antenna input and the receiver is operated in the AM AGC position. The audio output shall not vary more than 10 db with signal levels from 0.1 V to 0.5 V for the same conditions.

3.6.11 Beat frequency oscillator. The BFO shall oscillate at three crystal controlled frequencies, 453.3, 455.0 and 456.7 kHz, each ± 0.01% to provide optimum reception of upper or lower sideband signals during SSB operation. Selection of sideband is achieved with function switch placed in USB or LSB positions and the calibration frequency is selected with the function switch in CAL position.

3.6.12 Audio output.

3.6.12.1 Audio power output. The audio power output shall not be less than 20 mW with 18 Vdc applied to the internal battery supply terminals. When the supply voltage is reduced to 14 Vdc, the output shall not be less than 10 mW.

3.6.12.2 Audio harmonic distortion. With a 10 µV AM signal and output adjusted to 10 mW, audio harmonic distortion plus noise shall not exceed 10%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.12.3 Audio response. With 1000 cps as a reference level the overall audio response shall be as follows:

Audio frequency (Hertz)	Response (db)
300	-3 ± 2
1000	0
2500	-3 ± 2
7000	-16 or greater

3.6.12.4 Microphonics. There shall be no objectionable microphonic noises greater than 3 db above the receiver quiescent value. An objectionable microphonic noise is defines as an audible interference, sustained or repetitious, which can affect the intelligibility of an incoming signal for a period longer than 3 seconds.

3.6.13 Diode output.

3.6.13.1 Diode output level. With 1.2 uV AM signal 30% modulated at 1000 Hz applied to the 50 ohm antenna input, the diode output shall not be less than 2.0 V as measured across a 50,000 ohm resistive load. The diode output shall be negative with respect to ground. The input signal level shall be 2.0 uV for temperature and humidity requirements.

3.6.13.2 Diode output isolation. The diode output shall show no degradation (change greater than 5%) due to BFO leakage when tested in accordance with 4.8.2.

3.6.14 Power supplies. The receiver shall be capable of operating from internally contained dry batteries and an internal power converter operated from a 26 Vdc vehicular source or a 115/230 Vac, 47 to 400 Hz source, selected by a switch located on the rear of the receiver.

3.6.14.1 Input power requirement. The total power drain shall not exceed 1.0 W for internal battery operation, 2.5 W for vehicular operation and 5.5W for 115/230 Vac operation. The input power shall be measured with dial light off.

3.6.14.2 Power converter.

3.6.14.2.1 Voltage regulation. The output voltage of the power converter shall be regulated to 11.2 ± 0.2 Vdc over the full range of AC, battery and vehicular inputs as follows:

Input Frequency	Input Voltage
60 Hz	200 to 250 V and 90 to 130 V
400 Hz	200 to 250 V and 90 to 130 V
dc (vehicular)	22 to 32 V
dc (battery)	14 to 20 V

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.14.2.2 Output voltage ripple. The output voltage ripple of the receiver power supply shall not exceed 10 millivolts rms when measured at the diode output.

3.6.14.3 Battery supply. The receiver shall use twelve 1½ volt standard dry cells, type BA-30/U or BA-3030/U, as an internal battery supply. Batteries shall not be supplied with the equipment.

3.6.14.4 Solid state device protection. All solid state circuitry shall be protected from destructive transients originating in the vehicular supply under the worst case conditions described by MIL-STD-1275, or originating in any other supply. This protection shall be accomplished by the use of semiconductor devices specifically designed for transient suppression in this particular application.

3.6.14.5 Reverse polarity protection. The receiver shall be protected against accidental reverse polarity. It shall not be damaged by reverse polarity input.

3.6.14.6 Panel meter. A meter on the front panel of the receiver shall monitor the dry battery voltage level, the internal power converter output voltage level, and the carrier level. The meter shall have negligible effect on the metered circuits. Full scale deflection during AGC operation shall indicate approximately a 0.5 volt input signal. The scale markings shall be logarithmic. No more than 10 percent deflection shall be indicated with no signal input. During manual gain control operation, full scale deflection shall indicate the beginning of an overload condition for AM operation. The meter shall be marked to indicate the dry battery pack condition. The scale shall be marked with "good" and "replace", color coded red and green, and shall be calibrated such that the upper limit of the green area is 18 volts and the lower limit of this area is 14 volts.

3.6.15 Dial light. With all external power input options, the dial light shall operate when the dial light switch is in the ON or the M-ON position. Selection of the internal battery power input option shall limit dial light illumination to dial light switch position M-ON.

3.6.16 Final operation inspection. Immediately prior to preparation for delivery, each receiver shall be given a final operation inspection as specified in 4.8.2.

3.7 Environmental requirements. The equipment shall meet the following service conditions. Where a test is referenced, meeting the performance criteria shall be considered as compliance with the requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.1 Temperature.

3.7.1.1 Low temperature. The receiver shall be capable of operation without degradation at ambient temperatures as low as -40°F (-40°C) and of being stored at temperatures as low as -65°F (-54°C). Before, during and after testing as specified in 4.9.1.1, the receiver shall meet the following requirements:

Paragraph	Measurement	Performance
3.6.3.4	Freq Stab vs. Temp	As Specified
3.6.4.4	AM & CW Sensitivity vs. Temp	As Specified
3.6.5.2	Freq Calibrator Degradation	As Specified
3.6.5.3	Injection Level	As Specified
3.6.10	AGC Characteristic	As Specified
3.6.13	Diode Output	As Specified

3.7.1.2 High temperature. The receiver shall be capable of operation without degradation at ambient temperatures as high as + 154°F (68°C) and of being stored at temperatures as high as +160°F (71°C). Before, during and after testing as specified in 4.9.1.2 the receiver shall meet the same requirements as specified for low temperature.

3.7.2 Humidity. The receiver shall sustain no physical damage and shall be capable of operation at high temperature (86°F) and in high humidity (94%). Before, during and after testing as specified in 4.9.2 the receiver shall meet the performance specified in the tabulation below. After the humidity exposure, the receiver shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

Paragraph	Measurement	Performance
3.6.3.5	Freq Stab vs. Humidity	As Specified
3.6.4.5	AM & CW Sensitivity vs. Humidity	As Specified
3.6.5.2	Freq Cal Degradation	As Specified
3.6.5.3	Injection Level	As Specified
3.6.10	AGC Characteristic	As Specified
3.6.13	Diode Output	As Specified

Injection Level (3.6.5.3) Test will be conducted as a Prehumidity and Post Humidity Requirement only.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.3 Altitude. The receiver shall be capable of operation at 15,000 feet altitude and of being air transported at 40,000 feet. Before, during and after testing as specified in 4.9.3, the receiver shall meet full specified performance for the following measurements:

Paragraph	Measurement
3.6.5.1	Frequency Calibrator Accuracy
3.6.5.3	Injection Level

In addition, the receiver shall meet full specification performance for the following measurements before and after the testing specified in 4.9.3:

Paragraph	Measurement
3.6.4.1	AM & CW Sensitivity

3.7.4 Vibration and shock. The equipment shall be able to withstand the dynamic shock and vibration stresses, without degradation in performance or serious physical damage, normally encountered in servicing, in transportation by rail, air, sea, truck or semitrailer, or as loose truck cargo, or in vehicular use. The equipment shall meet the requirements of TABLE I before and after testing as specified in 4.9.4 through 4.9.9.

TABLE I - Vibration and shock requirements

Test	Test Para.	Performance after test	Notes
Vibration (Ground vehicle installation)	4.9.4	Specified performance No physical damage	A, E
Shock (basic design)	4.9.5	Specified performance No physical damage	A, E
Vibration (Secured cargo transportation)	4.9.6	Specified performance No physical damage	C, E
Shock (bench handling)	4.9.7	Specified performance No physical damage	D, F
Shock (loose cargo bounce)	4.9.8	Specified performance No physical damage, except for minor surface abrasions.	B, E
Shock (transit drop)	4.9.9	Specified performance Minor physical damage only, incl. inoperative panel meter	B, E

(See TABLE I (Cont'd) for Notes)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE I - Vibration and shock requirements (Cont'd)

NOTES:
A. Receiver shall be securely mounted on its shock mount, MT-4034/UR. Test applies to receiver/mount assembly.

B. Receiver tested in its canvas bag. The bag may be severely damaged in test.

C. Receiver under test, alone, is secured to vibration table.

D. Unit under test is receiver alone.

E. Specified performance for these tests is interpreted as meeting the following criteria:

Para. 3.6.4.1 - AM & CW Sensitivity at only one frequency in each band.
3.6.5.3 - Injection Level at only one frequency in each band.
3.6.13 - Diode Output at only one frequency in each band.

F. Specified performance for this test is interpreted as meeting the requirement of 3.6.4.1, CW sensitivity at one frequency only.

3.7.5 Rain. The equipment shall be capable of operation and shall show no evidence of physical damage or water leakage after exposure to rain accompanied with high wind as an anticipated service condition. Before and after testing as specified in 4.9.10, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM Sensitivity (AM only)
3.6.5.3	Injection Level

3.7.6 Immersion. Before and after testing as specified in 4.9.11, the receiver shall show no evidence of water leakage and shall meet full specified performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM Sensitivity (AM only)
3.6.5.3	Injection Level

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.7 Dust. The equipment shall be capable of operation and shall show no physical damage after exposure to fine blowing dust. Before and after testing as specified in 4.9.12, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM Sensitivity (AM only)
3.6.5.3	Injection Level

3.7.8 Salt fog. The equipment shall be capable of operation after exposure to marine or salt laden atmosphere. After testing as specified in 4.9.13, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. No rust shall be imposed on the test item by the test facility. The equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM Sensitivity (AM only) Injection Level

3.7.9 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9.14. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8 Electromagnetic interference characteristics. The receiver shall meet the electromagnetic interference requirements specified in MIL-STD-461A, Notice 4, Table A-1-MIL-E-55301, for Class CE equipment. Tests and test methods shall be as specified in MIL-STD-462, Notice 3, except that tuned channel frequencies shall be excluded for power line conducted susceptibility measurements of CS02. If the contractor considers it desireable to modify the receiver in any manner to facilitate conformance to the specified test limits, he shall obtain the approval of the contracting officer before proceeding with the modification.

3.9 Reliability: The reliability of the equipment shall be such that it is capable of meeting a lower test MTBF of 720 hours and an upper test MTBF of 2160 hours as defined in MIL-STD-781. During testing as specified in 4.11 the equipment shall meet the performance specified below:

a. Noise output of the receiver shall be constantly monitored during the length of the test. Cessation of noise output indicates failure.

b. In addition to the above, the receiver shall meet full specified performance for the following measurements:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the currentversion before use.

3.9 Reliability (Cont'd)

Paragraph	Measurement	Measurement Frequency
3.6.4.1	AM Sensitivity	Every second day } alternating
3.6.4.1	CW Sensitivity	Every second day } days
3.6.5.3	Injection Level	Daily
3.6.3.1	Freq Stability vs. Time	Every week } concurrent
3.6.5.1	Freq Calibrator Accuracy	Every week }
4.6.1	All Group A tests	Every third week

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268K(Army)

General requirements for plastic material and plastic parts
Wiring and Cabling, including:
  Slack
  Protection
  Clearance
  Splicing
Connections, general
Grounding
  Wire and cable shielding
Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging (see 4.4)).

b. Inspection covered by subsidiary documents (see 4.5).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.6).

(2) Quality conformance inspection of packaging (see 4.12).

4.3 Burn-in. Prior to the start of any or all of the following tests, each receiver shall be burned-in in accordance with SC-A-678020 and shall meet the requirements of that procedure and 3.3 herein.

4.4 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.4.1 First article units. Unless otherwise specified in the contract or purchase order, the contractor shall furnish four first article units of the complete equipment on order.

4.4.2 First article inspection. The first article inspection shall consist of inspections specified in TABLE II and shall be performed in the order specified herein.

4.4.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order (see 6.2.f).

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes; general	3.5.1
Finish	3.5.1.1
Marking	3.5.1.2
Safety	3.5.5
Human Engineering	3.5.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE II - First article inspection

Inspections	Req Para	Test Para	Number of Receiver Samples Three Units UNIT 1	UNIT 2	UNIT 3	Four Units UNIT 1	UNIT 2	UNIT 3	UNIT 4
1. Inspection covered by subsidiary documents.			Inspections to be performed on all units.						
2. Group A inspection	See Table III		Inspections to be performed on all units.						
3. Group B inspection	See Table IV		Inspections to be performed on all units.						
4. Group C inspection	As follows:								
Electrical			1			1			
Solid state device protection	3.6.14.4	4.8.2							
Reverse polarity	3.6.14.5	4.8.2	2			2			
Canvas shrinkage	3.5.13	4.9.15	3			3			
Tuning control	3.5.3	4.9.17	4			4			
EMI	3.8	4.9.18	5			5			
Environmental									
Temperature	3.7.1	4.9.1		1			1		
Humidity	3.7.2	4.9.2	6					1	
Altitude	3.7.3	4.9.3		2			2		
Vibration (Ground Vehicle)	3.7.4	4.9.4		5			4		
Shock (basic design)	3.7.4	4.9.5		6			5		
Vibration (secured cargo)	3.7.4	4.9.6			3				3
Bench handling	3.7.4	4.9.7			4				4
Loose cargo bounce	3.7.4	4.9.8		7				4	
Transit drop	3.7.4	4.9.9			5				5
Rain	3.7.5	4.9.10		3			3		
Immersion	3.7.6	4.9.11			1			2	
Dust (Note C)	3.7.7	4.9.12	7						1
Salt fog (Note C)	3.7.8	4.9.13		4					2
Fungus (Note C)	3.7.9	4.9.14			2			3	
5. Group D inspection Reliability	Receivers selected as required to meet 3.9 and 4.11								

Cont'd next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE II - First article inspection (Cont'd)

NOTES: a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspection 4 specify the order of inspections for the indicated unit. Subject to Government approval, the order of non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied at the contractors convenience.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.6 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5 and 4.6.1 through 4.6.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.6.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in TABLE III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using general inspection level II and the AQL's indicated in TABLE III.

4.6.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the final operation inspection (see 3.6.16 and 4.8.2) shall be last.

4.6.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. For the Group B interchangeability requirement, test samples may be separate units whose modules and replaceable parts were randomly selected and assembled prior to Group A testing. This inspection shall conform to TABLE IV and to the special procedures for small sample inspection of MIL-STD-105.

4.6.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in TABLE IV shall be S-4. All defects shall be considered major.

4.6.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE III - Group A inspection

Inspection	Req. Para.	Test Para.	AQL Major	AQL Minor
Visual and mechanical, General:	3.2, 3.5	4.5, 4.7		
Final assembly			1.0 dphu	4.0 dphu
Subassembly			1.0 dphu	4.0 dphu
Printed circuit boards			1.0%	4.0%
Modules			1.0%	4.0%
Accessory Kit MK-1517( )/UR			1.0 dphu	4.0 dphu
Maintenance Item Set			1.0 dphu	4.0 dphu
Electrical:				
Frequency range (overlap)	3.6.2	4.8.2		
AM sensitivity	3.6.4.1	4.8.2		
CW sensitivity	3.6.4.1	4.8.2		
Freq. calibrator accuracy	3.6.5.1	4.8.2		
Injection level	3.6.5.3	4.8.2		
Spurious indications	3.6.5.4	4.8.2		
Dial readout error	3.6.6	4.8.2		
Tuning backlash	3.6.7	4.8.2	1.5%	
AGC characteristic	3.6.10	4.8.2		
Beat freq. oscillator	3.6.11	4.8.2	for	
Audio power output	3.6.12.1	4.8.2		*
Audio harmonic distortion	3.6.12.2	4.8.2	the	
Diode output level	3.6.13.1	4.8.2		
Diode output isolation	3.6.13.2	4.8.2	group	
Input power requirements	3.6.14.1	4.8.2		
Voltage regulation	3.6.14.2.1	4.8.2		
Output voltage ripple	3.6.14.2.2	4.8.2		
Panel meter	3.6.14.6	4.8.2		
Dial light	3.6.15	4.8.2		
Final operation inspection	3.6.16	4.8.2		

*All electrical defects are considered major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE IV - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Interchangeability - Electrical	3.5.7	4.10.1	1.0%
- Mechanical	3.5.7	4.10.2	4.0%
Input impedance	3.6.1.1	4.8.2 4.8.2.1	
Antenna input isolation	3.6.1.2	4.8.2	
Freq. stability vs. time	3.6.3.1	4.8.2	
Freq. stability vs. int. battery voltage	3.6.3.2	4.8.2	
Freq. stability vs. rec'd signal level	3.6.3.3	4.8.2	
AM sensitivity (whip antenna)	3.6.4.2	4.8.2	
AM sensitivity vs. int. battery voltage	3.6.4.3	4.8.2	4.0
Calibration adjustment	3.6.5.5	4.8.2	dphu
Image rejection	3.6.8.1	4.8.2	for
IF rejection	3.6.8.2	4.8.2	the
Crossmodulation	3.6.8.3	4.8.2	group
Intermodulation distortion products	3.6.8.4	4.8.2	
IF center frequency	3.6.9.1	4.8.2	
IF selectivity	3.6.9.2	4.8.2	
Audio response	3.6.12.3	4.8.2	
Microphonics	3.6.12.4	4.8.2	
Battery supply	3.6.14.3	4.8.2	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.6.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in TABLE V. Samples shall be selected in accordance with 4.6.3.1.

4.6.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup 1 through and including Subgroup 4 in TABLE V. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup 1 through Subgroup 4 in TABLE V from every 75 units and fraction thereof produced. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

4.6.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government; see NOTES a and c of TABLE II.

4.6.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.6.3.4 Reinspection of conforming Group C sample units. Unless otherwise specified sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspection.

4.6.4 Group D inspection. This inspection shall consist of the tests specified in TABLE VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods and procedures shall be in accordance with paragraph 4.11. Individual units which have been subject to potentially damaging environmental tests will not be selected for reliability testing. Corrective action as a result of a failed Group C inspection may require that this inspection be reconducted.

4.6.4.1 Group D inspection lot formation. The inspection lot for production reliability acceptance testing shall be 100 units or fraction thereof produced, except that a final quantity of 42 or less may be added to a last lot of 100 if the procurement quantity is not an even multiple of 100.

4.6.4.2 Sampling for Group D inspection. The test sample for production reliability acceptance shall be seven (7) to ten (10) percent of the inspection lot, but not less than 3 units. It shall be randomly selected from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE V - Group C inspection

Inspection	Req Par	Insp Par
Subgroup 1		
Solid state device protection	3.6.14.4	4.8.2
Reverse polarity protection	3.6.14.5	4.8.2
Canvas shrinkage	3.5.1.3	4.9.15
Size and weight	3.5.2	4.9.16
Tuning control	3.5.3	4.9.17
Electromagnetic interference	3.8	4.9.18
Subgroup 2		
Temperature	3.7.1	4.9.1
Altitude	3.7.3	4.9.3
Rain	3.7.5	4.9.10
Vibration (ground vehicle)	3.7.4	4.9.4
Shock (basic design)	3.7.4	4.9.5
Subgroup 3		
Humidity	3.7.2	4.9.2
Immersion	3.7.6	4.9.11
Fungus (NOTE 1)	3.7.9	4.9.14
Loose cargo bounce	4.7.4	4.9.8
Subgroup 4		
Dust (NOTE 1)	3.7.7	4.9.12
Salt fog (NOTE 1)	3.7.8	4.9.13
Vibration (secured cargo)	3.7.4	4.9.6
Bench handling	3.7.4	4.9.7
Transit drop	3.7.4	4.9.9

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

TABLE VI - Group D inspection

Inspection	Req Para	Test Para
Reliability	3.9	4.11

4.6.4.3 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.6.4.4 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the currentversion before use.

4.7 Visual and mechanical inspection. The equipment shall be visually and mechanically inspected to insure conformance to the construction requirements of 3.2, 3.5 and the workmanship requirements of 3.10. Defects shall be classified in accordance with MIL-STD-252(EL).

4.8 Electrical tests.

4.8.1 Standard test conditions. The electrical tests shall be conducted under the following conditions:

a. Unless otherwise specified, all tests shall be conducted under conditions of normal room ambient temperature and humidity.

b. Unless otherwise specified, all tests shall be conducted using the nominal 115V input, however the various rated input voltages are as follows:

(1)	Vehicular supply	26 ± 0.5 Volt dc
(2)	115 Volt AC supply	115 ± 2V, 60 Hz
(3)	230 Volt AC supply	230 ± 4V, 60 Hz
(4)	Dry battery supply	18 Volts ± 0.5 Volt dc

c. RF input - signal generator with 50 ohm output impedance using a 50 ohm pad in series.

d. Modulation

AM - - - - - 30% at 1000 Hz

e. Audio output load impedance - - - 600 ohms

f. Audio output level 10 mW, unless otherwise specified.

4.8.2 Detailed electrical tests. The equipment shall meet the applicable requirements of Section 3 of this specification when tested in accordance with the procedures of SM-A-747531 as identified in TABLE VI, below:

4.8.2.1 Input impedance. In addition to the test described by paragraph 6.6 of SM-A-747531, measure input impedance with a spectrum analyzer, sweeping the entire frequency range.

4.8.2.2 Battery supply. In addition to the tests described by SM-A-747531, read the voltage available at dust cover J1, pin 4 when 12 fresh BA-30/U or BA-3030/U cells are in place. It shall be $\geq$ 18 Vdc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE VI - Performance tests

Test Title	Req. Par.	SM-A-747531 Par. No.	Test Group
Input impedance	3.6.1.1	6.6 of SM-A-7747531 and 4.8.2.1 of this spec.	B
Antenna input isolation	3.6.1.2	6.6	B
Frequency range (overlap)	3.6.2	5.6	A
Freq. stability vs. time	3.6.3.1	6.7	B
Freq. stability vs. internal battery voltage	3.6.3.2	6.8	B
Freq. stability vs. received signal level	3.6.3.3	6.9	B
Freq. stability vs. temperature	3.6.3.4	7.8.1	C
Freq. stability vs. humidity	3.6.3.5	7.9.2	C
AM sensitivity	3.6.4.1	5.7	A
CW sensitivity	3.6.4.1	5.8	A
AM whip antenna sensitivity	3.6.4.2	6.10	B
AM sensitivity vs. internal battery voltage	3.6.4.3	6.11	B
AM & CW sensitivity vs. temperature	3.6.4.4	7.8.3	C
AM & CW sensitivity vs. humidity	3.6.4.5	7.9.3	C
Freq. calibrator accuracy	3.6.5.1	5.9	A
Freq. calibrator degradation	3.6.5.2	7.8.4	C
Injection level	3.6.5.3	5.10	A
Spurious indications	3.6.5.4	5.11	A
Calibration adjustment	3.6.5.5	6.5	B
Dial readout error	3.6.6	5.12	A
Tuning backlash	3.6.7	5.4	A
Image rejection	3.6.8.1	6.12	B
IF rejection	3.6.8.2	6.13	B
Crossmodulation	3.6.8.3	6.14	B
Intermodulation distortion prod.	3.6.8.4	6.15	B
IF center frequency	3.6.9.1	6.16	B
IF selectivity	3.6.9.2	6.17	B
AGC characteristics	3.6.10	5.13	A
Beat frequency oscillator	3.6.11	5.14	A
Audio power output	3.6.12.1	5.15	A
Audio harmonic distortion	3.6.12.2	5.16	A
Audio response	3.6.12.3	6.18	B
Microphonics	3.6.12.4	6.19	B
Diode output level	3.6.13.1	5.17	A
Diode output isolation	3.6.13.2	5.18	A
Input power requirements	3.6.14.1	5.19	A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE VI - Performance tests (Cont'd)

Test Title	Req. Par.	SM-A-747531 Par. No.	Test Group
Voltage regulation	3.6.14.2.1	5.20	A
Output voltage ripple	3.6.14.2.2	5.21	A
Battery supply	3.6.14.3	4.8.2.2 of spec.	B
Solid state device protection	3.6.14.4	7.7.1	C
Reverse polarity protection	3.6.14.5	7.7.2	C
Panel meter	3.6.14.6	5.3	A
Dial light	3.6.15	5.5	A
Final operation	3.6.16	5.22	A

4.9 Environmental tests. All environmental test methods, conditions and procedures given below are contained in MIL-STD-810C. The equipment performance measurements shall be made under standard conditions as defined in MIL-STD-810C. Successful completion of any particular test shall be considered evidence of meeting the corresponding service condition.

4.9.1 Temperature. The equipment shall be tested according to the following test methods:

4.9.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810C and shall meet the requirements of 3.7.1.1.

Storage temperature (Step 2)	-65°F (-54°C)
Operating temperature (Steps 4, 5)	-40°F (-40°C)
Make measurements at Steps 1, 5, 7	

4.9.1.2 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II of MIL-STD-810C and shall meet the requirements of 3.7.1.2.

Cyclic Stress Temp. (Steps 2, 3, 5)	+ 120°F (+49°C)
Cyclic Stress Temp. (Step 4)	+160°F (+71°C)
Operating temperature (Steps 7, 8)	+154°F (+68°C)
Make measurements at Steps 1, 8, 10	

4.9.2 Humidity. The equipment shall be tested in accordance with Method 507.1, Procedure III of MIL-STD-810C. In Step 7, the chassis shall be removed from the dust cover so as to expose the normally sealed areas to the chamber environment. The equipment shall be operated during the last 5 hours of the last cycle of Step 6 and shall meet the requirements of 3.7.2 when tested in Steps 4, 6 (last cycle only), 8 and 11.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:16Z
Check the source to verify that this is the current version before use.

4.9.3 Altitude. The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810C. A non-operating test shall be performed for 1 hour at 40,000 feet prior to the operating test. The equipment shall meet the requirements of 3.7.3 when tested at Steps 1, 2 and 4.

Operating altitude	15,000 feet
Non-operating altitude	40,000 feet

4.9.4 Vibration (ground vehicle installation). The receiver, when mounted on its Shock Mount Base, MT-4034/UR, shall be tested in accordance with Method 514.2, Procedure VIII of MIL-STD-810C and shall meet the requirements of 3.7.4.

Equipment category	F
Applicable table	514.2 - VI
Applicable figure	514.2 - 6
Test curve	V (1.5g)
Sweep	12 min., 5-200-5Hz
Cycle time	3 hrs. each, 3 directions

4.9.5 Shock (basic design). The receiver, when mounted on its Shock Mount Base, MT-4034/UR, shall be tested in accordance with Method 516.2, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.4.

Equipment category	ground vehicle
Applicable figure	516.2-1
Peak shock value	40 g
Duration	11 ms

4.9.6 Vibration (secured cargo transportation). The receiver alone, without Shock Mount, shall be tested in accordance with Method 514.2, Procedure X of MIL-STD-810C and shall meet the requirements of 3.7.4.

Transport mode	rail, air, sea, truck or semi
Applicable table	514.2 - VII
Applicable figure	514.2-7
Test curve	AW (1.5 g)
Sweep	12 min., 5-200-5Hz
Cycle time	84 min. each, 3 axes

4.9.7 Shock (bench handling). The equipment shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810C, and shall meet the requirements of 3.7.4.

NON-
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.9.8 Shock (loose cargo bounce). The equipment shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810C, and shall meet the requirements of 3.7.4.

4.9.9 Shock (transit drop). The receiver,in the canvas bag, shall be tested in accordance with Method 516.2, Procedure II, of MIL-STD-810C, except that the drop height shall be 30 inches. Receiver only shall be in canvas bag during test. Prior to and after testing the equipment shall meet the requirements of 3.7.4.

4.9.10 Rain. The receiver,excluding the canvas bag, shall be tested in accordance with Method 506.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.5.

4.9.11 Immersion. The receiver, excluding the canvas bag, shall be tested in accordance with Method 512.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.6.

4.9.12 Dust. The receiver, excluding the canvas bag,shall be tested in accordance with Method 510.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.7.

4.9.13 Salt fog. The equipment, including the receiver and all accessories (the receiver not in canvas bag during salt atmosphere exposure), shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.8.

4.9.14 Fungus. The equipment in the assembled and as ready for delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810C, Method 508.1, Procedure I, except that the temperature humidity conditions of paragraph 3.1.6a shall be changed to a constant 29°C (84°F) and 95% RH (minimum) during the life of the test. The equipment or samples shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 28 days, based on visual examination, the equipment assembly or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it does not meet criteria of 3.7.9 or if it shows more than the growth specified above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.9.15 Canvas shrinkage. The canvas bag shall meet the requirements of 3.5.1.3 before and after testing in accordance with SM-A-747531 paragraph 7.3.

4.9.16 Size and weight. When measured and weighed in accordance with SM-A-747531 paragraph 7.4, the receiver shall meet the requirements of 3.5.2.

4.9.17 Tuning control. The tuning control of the receiver shall meet the requirements of 3.5.3 when tested in accordance with SM-A-747531 paragraph 7.5.

4.9.18 Electromagnetic interference. The equipment shall be tested in accordance with the test setups and measurement techniques of MIL-STD-462, Notice 3 as implemented by a contractor prepared Government approved EMI test plan and shall meet the requirements of 3.8. Exclude tuned channel frequencies for power line conducted susceptibility measurements of CS02.

4.10 Interchangeability. The equipment shall be inspected to insure the physical and functional interchangeability requirements of 3.5.7.

4.10.1 Electrical interchangeability. All like modules and replaceable parts shall be tested for functional interchangeability by assembly of randomly selected parts. Following the random assembly process, the equipment shall be tested to and shall meet Group A acceptance requirements. Electrical retuning or realignment of modules during production assembly may be expected and are permissible. Random selection for this assembly shall be supervised by the Government.

4.10.2 Mechanical interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.5.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

f. Compatibility of switching on RF circuit boards and shaft.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:16Z
Check the source to verify that this is the current version before use.

4.11 Reliability. The equipment shall be tested in accordance with test plan VC of MIL-STD-781 to demonstrate a lower test MTBF of 720 hours and shall meet the requirements of 3.9. The environmental conditions and duty cycle shall be in accordance with SC-A-678020.

4.12 Quality conformance inspection for packaging. Quality conformance inspection for packaging shall be in accordance with Group A, Group B and, when required (see 6.2.c), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.13 Rough handling test (packaging). When the rough handling test of specification MIL-P-116 is required by the contract (see 6.2.c), the AM sensitivity and diode output requirements of 3.6.4.1 and 3.6.13 shall be met to determine freedom from operational malfunction caused by rough handling.

5. PACKAGING.

5.1 Packaging. Packaging shall be in accordance with SPI 1G00075.

6. NOTES.

6.1 Intended use. The AN/URR-70 receiving set is an HF superheterodyne set capable of receiving AM, CW and SSB signals in five bands over a frequency range of 0.536 to 20.5 MHz. While it is capable of fixed station, mobile or manpack use, its primary intended uses include direction finding in connection with the AN/TRQ-30, AN/TRQ-32 and AN/TRR-33. The flat top, relatively vertical side IF skirt makes this receiver ideal for DF work.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or Level B preservation and packing (see section 5).

c. When rough handling test is required (see 4.13).

d. First article inspection: Four sample units of each item cited in section 1 are generally required to reduce the overall time for lengthy environmental and performance tests (may be modified by contract). Only two samples are required for the Accessory Kit MK-1517/UR.

e. Marking and shipping of samples.

f. Data requirements. Deliverable data required by this specification is cited in 4.4.3 entitled First Article Data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

6.2 Ordering data (Cont'd)

g. Place of final inspection.

h. Technical literature required.

i. Quantity of tools and running spare parts required.

6.3 Location of operational inspection. It is desirable that the operational inspection (3.6.16/4.8.2) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 Color. The color chip furnished by the procuring agency (see 3.5.1.1), will be color chip No. X-24087, and may be obtained upon request to Commanding Officer, U.S. Naval Depot, 5801 Tabor Avenue, Philadelphia, PA.

6.5 Color of textiles (see 3.5.1.3.1). Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.6 Definitions.

6.6.1 Temperature stabilization. Temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag does not change more than 2.0°C (3.6°F) per hour.

6.6.2 Physical damage. Physical damage is defined as corrosion, rust, blistering, swelling or other deterioration of parts and materials.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN

Army-CR

PREPARING ACTIVITY
Army - CR
Proj. No. 5820-A684

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

MIL-R-55317A(CR)

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-55317A(CR) Receiving Set, Radio AN/URR-70, and Kit, Accessory MK-1517/UR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

HQS, USA CECOM
ATTN: DRSEL-ED-TO
FORT MONMOUTH, NJ 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.